1. Overview of Machine Learning Paradigms

Machine learning is a subfield of artificial intelligence concerned with algorithms that improve their performance metric P on a specified class of tasks T through empirical experience E.

Taxonomy of Machine Learning:
1. Supervised Learning: The algorithm is presented with an annotated training dataset $D = \{(x_i, y_i)\}$ for $i = 1$ to N, where $x_i \in \mathbb{R}^d$ represents input feature vectors and y_i represents ground truth labels.
   - Classification: Discrete target labels $y \in \{1, 2, ..., C\}$.
   - Regression: Continuous real-valued targets $y \in \mathbb{R}$.

2. Unsupervised Learning: The algorithm receives unlabeled inputs $X = \{x_1, ..., x_N\}$ and seeks to discover latent structures, density distributions, or low-dimensional manifolds. Exemplified by K-Means clustering, Gaussian Mixture Models, and Principal Component Analysis (PCA).

3. Reinforcement Learning: An agent interacts with a stochastic environment through a Markov Decision Process (MDP), taking actions to maximize cumulative expected discounted reward over time.

2. Overfitting, Underfitting, and the Bias-Variance Tradeoff

The ultimate goal of supervised machine learning is out-of-sample generalization rather than mere training set memorization.

The expected generalization error of a predictive model decomposes mathematically into three distinct components:
E[(y - f_hat(x))^2] = Bias[f_hat(x)]^2 + Var[f_hat(x)] + sigma^2

Definitions:
- Bias: The systemic error introduced by approximating a complicated real-world function with a simplified model family. High bias results in underfitting.
- Variance: The sensitivity of the model to specific idiosyncratic fluctuations in the training dataset. High variance results in overfitting.
- Irreducible Error (sigma^2): The inherent noise in the data generating process that cannot be eliminated by any modeling technique.

Regularization Techniques to Prevent Overfitting:
- L2 Regularization (Ridge / Weight Decay): Penalizes large weights by adding (lambda/2) * ||w||_2^2 to the loss function, encouraging smooth, diffuse weights.
- L1 Regularization (Lasso): Adds lambda * ||w||_1 to the loss function, inducing sparsity by driving non-informative coefficients exactly to zero.
- Dropout: Randomly deactivates neurons during training with probability p, preventing co-adaptation of features.

3. Optimization via Gradient Descent

Most modern statistical learning algorithms minimize an empirical risk objective function J(theta) over parameter space theta in R^p:
J(theta) = (1/N) * sum_{i=1}^N L(f(x_i; theta), y_i)

Gradient Descent Variants:
1. Batch Gradient Descent: Computes the exact gradient nabla_theta J(theta) using the entire training dataset at each step:
theta_{t+1} = theta_t - eta * nabla_theta J(theta_t)
Guarantees convergence to global minimum for convex functions, but computationally prohibitive for large datasets.

2. Stochastic Gradient Descent (SGD): Computes an unbiased gradient estimate using a single randomly chosen training instance at each update step:
theta_{t+1} = theta_t - eta * nabla_theta L(f(x_i; theta_t), y_i)

3. Mini-Batch Gradient Descent: Evaluates gradients across small batches of B samples (typically B in {32, 64, 128, 256}), achieving numerical stability through vectorized matrix operations on GPUs while retaining stochastic regularizing noise.

4. Deep Neural Networks and Backpropagation

Multilayer Perceptrons (MLPs) are universal function approximators composed of stacked affine transformations interleaved with nonlinear element-wise activation functions.

For layer l with weights W^(l) and bias b^(l):
z^(l) = W^(l) a^(l-1) + b^(l)
a^(l) = sigma(z^(l))

Common Activation Functions:
- Rectified Linear Unit (ReLU): f(z) = max(0, z). Mitigates the vanishing gradient problem in deep architectures.
- Leaky ReLU: f(z) = max(alpha * z, z) where alpha = 0.01, preventing dead neurons.
- Sigmoid: f(z) = 1 / (1 + exp(-z)). Maps activations to interval (0, 1), primarily utilized in binary classification output layers.

The Backpropagation algorithm computes partial derivatives partial L / partial W^(l) via recursive application of the multivariate chain rule from the final layer backward to the input layer.

5. Performance Evaluation Metrics for Academic Research

Evaluating statistical classifiers requires rigorous metrics beyond simple accuracy, particularly in datasets characterized by severe class imbalance.

Confusion Matrix Definitions:
- True Positives (TP): Correctly identified positive instances.
- False Positives (FP): Negative instances incorrectly classified as positive (Type I error).
- False Negatives (FN): Positive instances incorrectly classified as negative (Type II error).
- True Negatives (TN): Correctly identified negative instances.

Core Equations:
- Precision: TP / (TP + FP). Reflects the accuracy of positive predictions.
- Recall (Sensitivity): TP / (TP + FN). Reflects the model's ability to locate all actual positives.
- F1-Score: The harmonic mean of Precision and Recall:
  F1 = 2 * (Precision * Recall) / (Precision + Recall)
- Area Under the ROC Curve (AUC-ROC): Evaluates the true positive rate against the false positive rate across all discrimination thresholds.